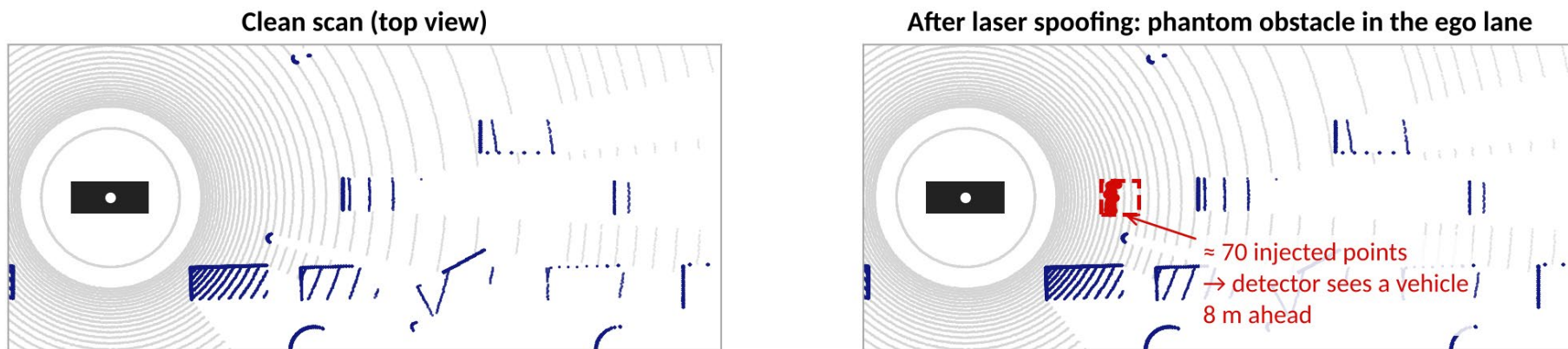


L4.2 Adversarial Attacks for AD



Zonghua Gu | Fall 2026

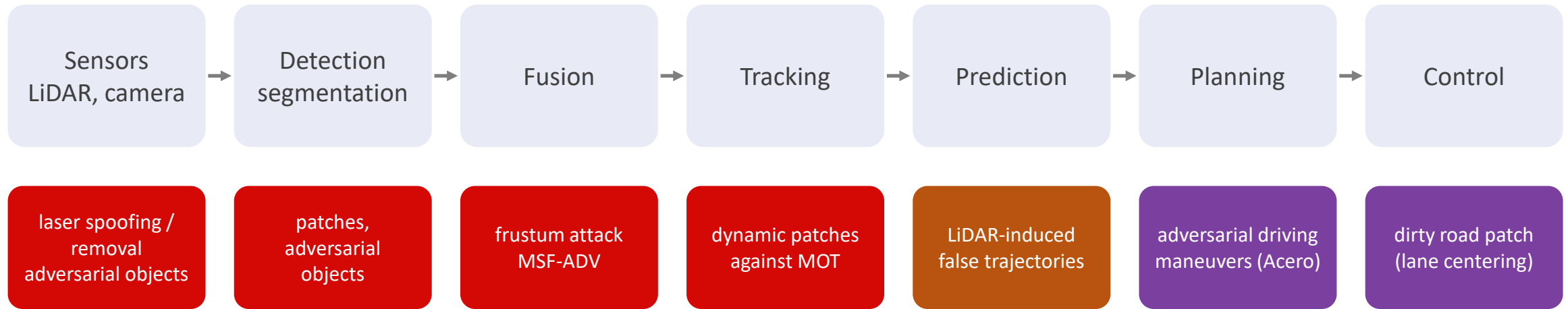
Outline

- Recap and the AD attack surface
- Attacks on object tracking
- Attacks on LiDAR perception
- Is sensor fusion safe?
- From perception errors to driving consequences
- Defenses and discussion

Recap: the L3.2 Toolbox

L3.2 concept	What we learned	What changes for autonomous driving
Perturbation budget	Small ℓ_p -norm δ keeps the input 'the same' for a human	Physics replaces the norm ball: the attack must be printable, projectable, or a placeable object
White-box gradients (FGSM, PGD)	Differentiate the loss w.r.t. the input	Pipeline stages (NMS, voxelization, tracking) are non-differentiable: need proxies or search
Physical attacks	Optimize over viewpoints, lighting, and spatial masks	Must also survive distance, motion, sensor noise, and time
Black-box transfer	Craft on a proxy model, hit the target	Open AD stacks (Apollo, Autoware, OpenPilot) are realistic white-box targets
Adversarial training	min-max training raises robustness	Costs accuracy and real-time budget; does not cover non-image attack surfaces

The Whole Driving Loop Is the Target



attack entry points covered today

- Errors injected early may be **filtered** (a tracker smooths one bad frame) or **amplified** (a wrong lane center steers the car)
- The attacker chooses the weakest layer, not the one we trained adversarially in L3.2
- A fair security evaluation therefore measures the **outcome at the controller**, not only model accuracy

Threat Model Dimensions for AD Attacks

Dimension	Options	AD examples in this lecture
Attack channel	digital · physical sensor (laser, projector) · physical object · V2X message · driving behavior	laser spoofing; adversarial 3D-printed object; fabricated V2X features; Acero maneuvers
Attacker knowledge	white-box · gray-box · black-box	white-box on open stacks (Apollo, Autoware); black-box spoofing (Sun et al.)
Goal	hide an object (false negative) · create a phantom (false positive) · mislead class or trajectory · degrade tracking or control	vehicle hiding; phantom obstacle; wrong predicted trajectory; lane departure
Constraints	stealth · cost · physical realizability · timing (attack must be on-line)	70 spoofed points; 0.3 m objects; patch on a road; normal-looking driving

Three Levels of Attack Success

Level 1

Model-output error

A frame is misclassified, an object is missed, or a phantom appears
metric: attack success rate, mAP drop

Level 2

Persistent perception failure

The error survives tracking and fusion over many frames
metric: track loss, ID switches, duration of failure

Level 3

Closed-loop safety violation

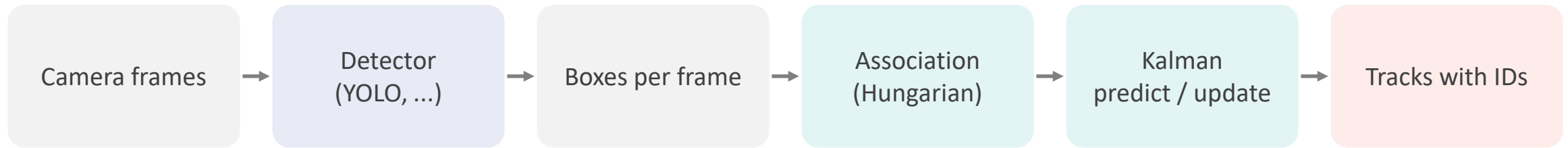
The vehicle collides, leaves its lane, or brakes hard
metric: collision rate, time to violation, rule violations

stronger claims need more evidence: level 1 does not imply level 2, and level 2 does not imply level 3

Outline

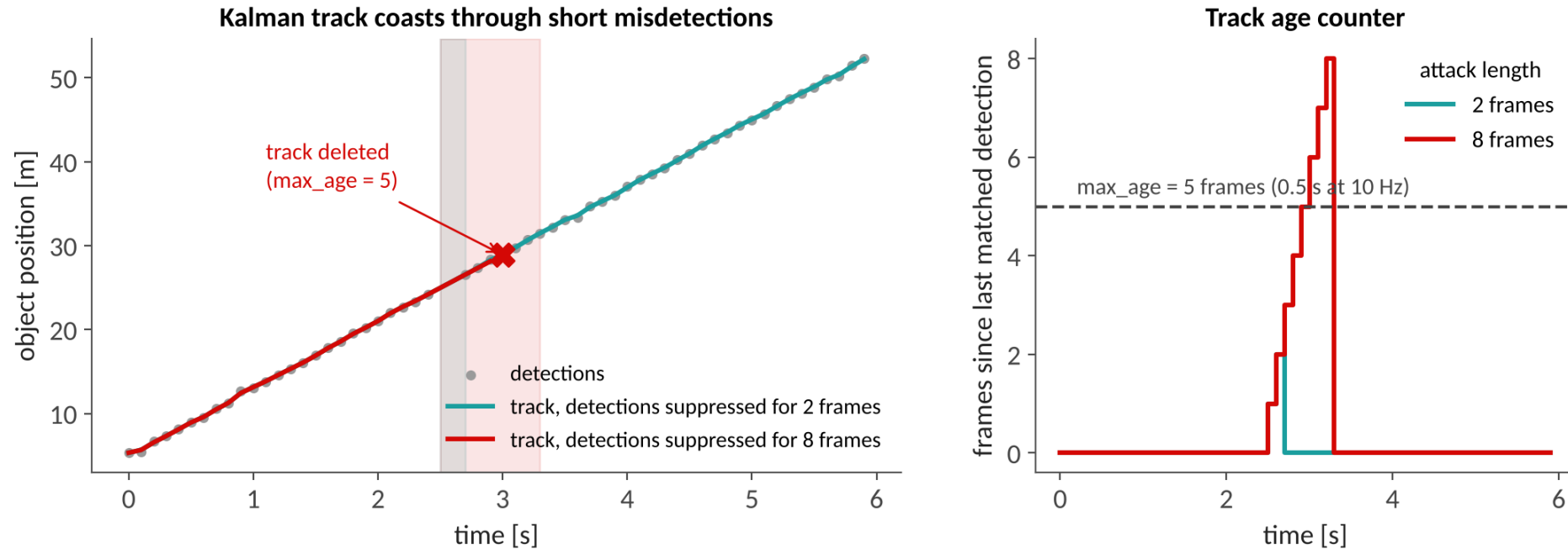
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Detection vs. Multi-Object Tracking



- Planning consumes **tracks** (identity, velocity), not raw single-frame detections
- Each track is **predicted** forward with a motion model (e.g., constant velocity) and corrected by matched detections
- A track that is unmatched for more than **max_age** frames is deleted; new tracks are created from unmatched detections
- So the tracker acts as a **temporal low-pass filter**: isolated misses are bridged by prediction
- Consequence for the attacker: fooling the detector in one frame is not enough

Why Tracking Changes the Attack



- Toy model: constant-velocity object, Kalman tracker, 10 Hz, max_age = 5 frames. Suppressing detections for **2 frames** changes nothing; **8 frames** deletes the track
- Real finding: detection attacks alone are often not enough: over **98 %** detection-attack success may be needed to change tracking outcomes (Jia et al., ICLR 2020)
- Attack design must therefore reason about **association and track management**, not just boxes

Attacking MOT: Threat Model (WIP)

- Chen Ma, Ningfei Wang, Qi Alfred Chen, Chao Shen: Towards the Practicality of the Adversarial Attack on Object Tracking in AD
- **Gap:** prior tracking attacks target a single-object tracker (Muller et al., CCS 2022) or are digital only (Jia et al., ICLR 2020); none attacks **multi-object tracking in the physical world**
- **Threat model:** white-box access to the perception pipeline; **dynamic adversarial patches** shown on a monitor or projector
- **Goal:** induce tracking errors that lead to a crash or an emergency stop
- Why dynamic? A static patch cannot adapt to the tracker's state frame by frame

Method Sketch: Beating NMS and the Tracker

1 Select boxes

Choose which bounding boxes the patch should attack, so that the tracker's association is steered, not just one detection removed

2 Handle NMS

Detector outputs pass through non-maximum suppression; the patch must shape boxes so that the intended ones survive

3 Balance two losses

Shape / position loss and score loss conflict; a dual-loss optimization resolves the conflict

- The attack optimizes over the **detector + NMS + tracker** pipeline, not over a classifier alone
- Non-differentiable steps (NMS, association) must be handled by design of the loss

Results and How to Quote Them

90 %

success rate
YOLOv3 (Autoware.AI)

80 %

success rate
Apollo model

10

Berkeley Deep Drive
video clips

What this shows

A patch optimized against detector + tracker can change tracking outcomes in a (digital / preliminary) evaluation: level 1-2 evidence

What it does not show

General risk to deployed AVs; robustness of the physical patch; closed-loop crashes (future work per the authors: large-scale evaluation, physical robustness)

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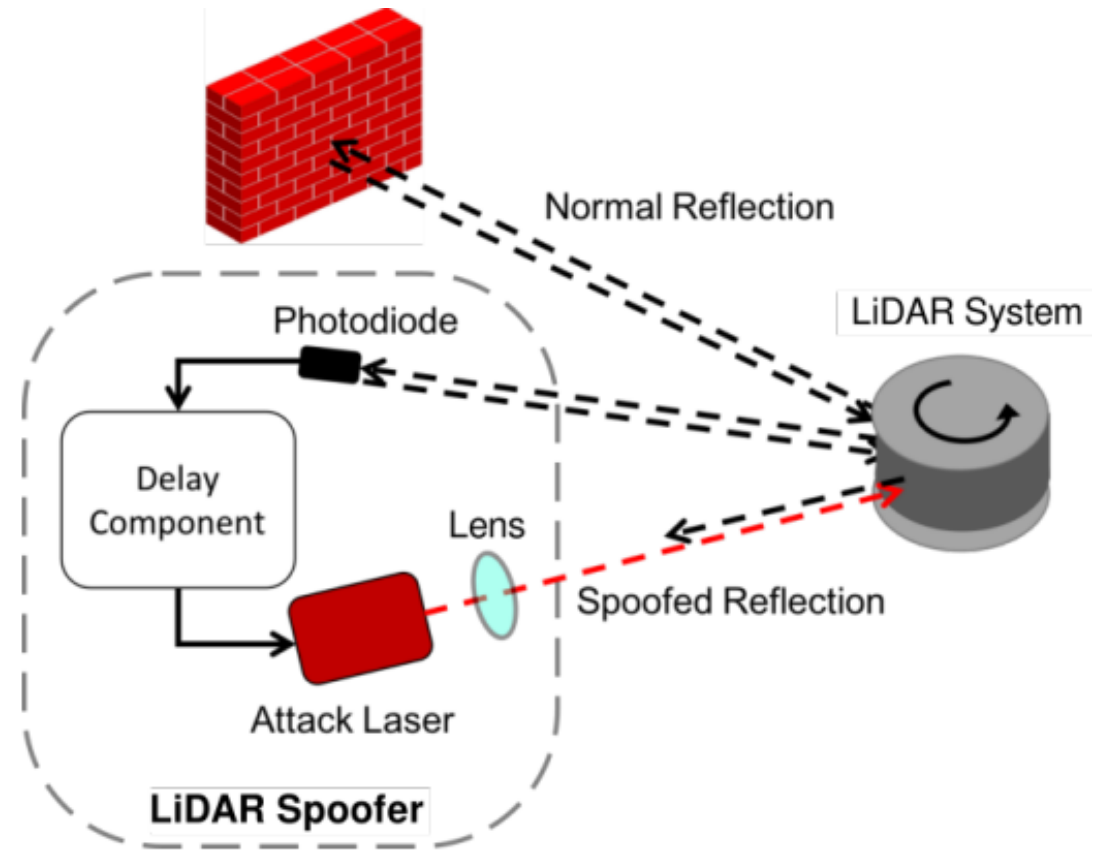
LiDAR Attack Surface

Channel	Mechanism	Typical goal	Examples
Laser	inject returns that the sensor cannot distinguish from real ones, or blind / saturate it	phantom obstacle; remove an obstacle	Cao et al. CCS 2019; Sun et al. USENIX Sec 2020; Cao et al. USENIX Sec 2023
Objects in the scene	place reflective objects whose geometry fools the network	hide a vehicle; change road surface; false trajectory	Tu et al. CVPR 2020; Zhu et al. SenSys 2021; Lou et al. USENIX Sec 2024
Network (V2X)	send fabricated sensor data or features to other vehicles	corrupt shared perception	Zhang et al. USENIX Sec 2024

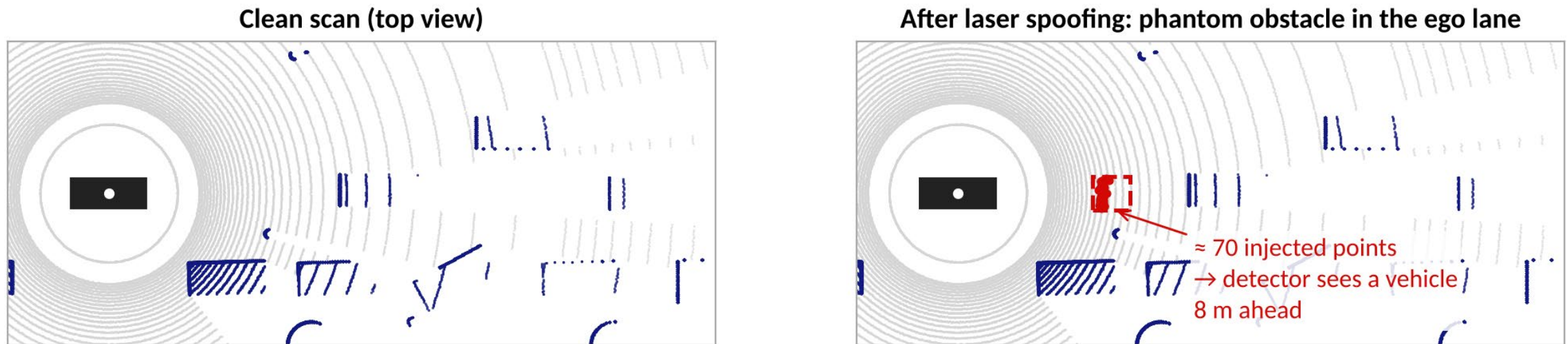
LiDAR measures $R = c \cdot \Delta t / 2$ and trusts any return with the right wavelength and timing: it cannot tell a real reflection from an injected one.

Laser Spoofing: Injecting Phantom Points

- A **photodiode** detects the LiDAR's outgoing pulse, a **delay component** waits, and an **attack laser** fires a spoofed return
- The delay sets the apparent range: points appear at a chosen distance, e.g., in the ego lane
- The attacker does not break the sensor; it exploits that the sensor trusts any matching return
- Naive attacks injected a few points; modern detectors discard them as noise (next slide)

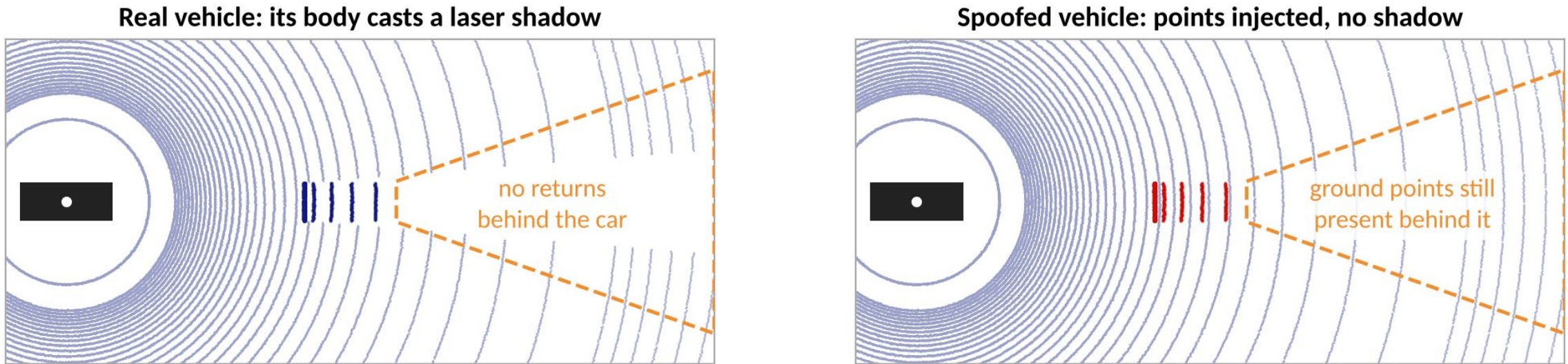


Naive Spoofing Is Not Enough (CCS 2019)



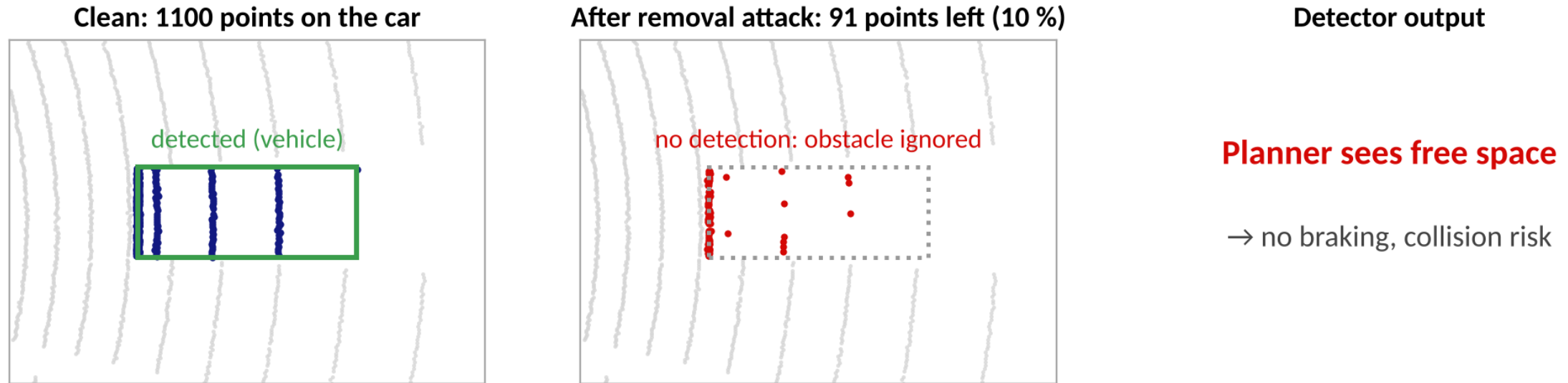
- First security study of **LiDAR-based perception in AD**; blind spoofing is insufficient because the ML detector filters implausible points
- Fix: **optimization-based attack** combined with global sampling to find point patterns the detector accepts as a vehicle: about **75 %** success rate
- Two real-world attack scenarios show impact on **safety and mobility**; defenses are proposed at AV-system, sensor and model levels
- Left: clean synthetic scan; right: ≈ 70 injected points forming a phantom vehicle 8 m ahead (our illustration)

Black-Box Spoofing and the Occlusion Pattern



- Sun et al. (USENIX Security 2020): perception models overlook **occlusion patterns**; a general black-box spoofing attack reaches $\approx 80\%$ mean success across target models
- A real vehicle blocks the laser: no returns behind it. Spoofed points leave the space behind them populated
- Defenses: **CARLO** (detect violations of the occlusion pattern) lowers success to $\approx 5.5\%$; **SVF** (a robust architecture) to $\approx 2.3\%$

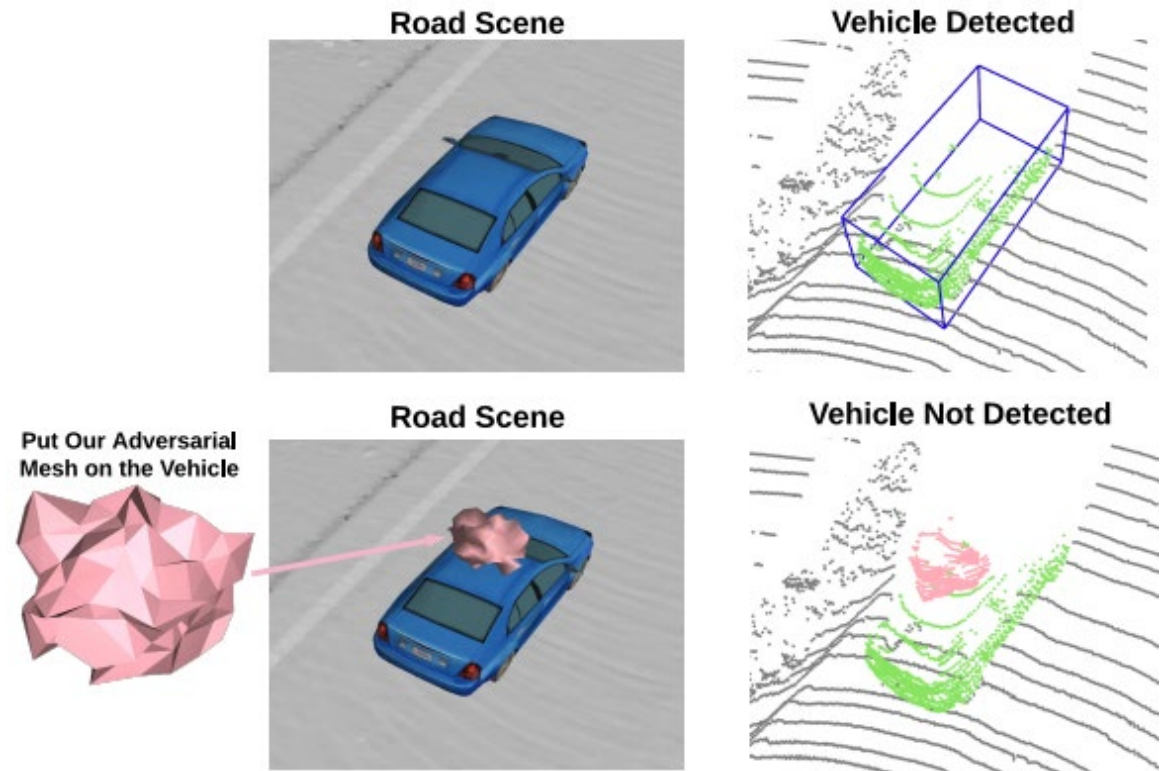
Removal Attacks: You Can't See Me



- Cao et al. (USENIX Security 2023): **physical removal attack** eliminates point-cloud returns of a target at the sensor, before detection
- **92.7 %** success removing **90 %** of a target's points on moving vehicles, even at 45° angles; invisible to the human eye
- Evaluated on Apollo, Autoware, PointPillars and three fusion models (Frustum-ConvNet, AVOD, Integrated-Semantic-Level Fusion)
- Exploits sensor-level filtering and transformation, so a robust detector never sees the object

Adversarial Objects: Hiding a Car from LiDAR

- Tu et al. (CVPR 2020): a universal **3D adversarial mesh** placed on the **roof** of a vehicle hides it from LiDAR detectors
- **80 %** success rate in hiding the vehicle
- Tested against multiple detectors with different point-cloud representations; data augmentation was explored as a preliminary defense
- A physical object, not a signal: no special hardware at attack time



Attacking LiDAR Segmentation (SenSys 2021)

- First **physically realizable** attack on LiDAR **point-cloud semantic segmentation**
- Two goals: **vehicle hiding** (parked car disappears) and **road surface changing** (road labeled as vegetation)
- Adversarial objects: clusters of at most **10 points**, up to **0.3 m**, any shape that reflects LiDAR: cardboard, traffic signs, billboards
- Models: PointNet, PointNet++, PointASNL, Cylinder3D, SqueezeSeg; data: SemanticKITTI plus real Ouster OS1-64 scans
- Reported **> 90 %** success in real-world driving environments; robust to ≈ 0.3 m placement error (**> 50 %** effective)

The Attack as an Optimization Problem

min over cluster positions s : $L(s) = L_{\text{seg}} + \alpha \cdot L_{\text{sem}} + \beta \cdot L_{\text{occ}}$, $\alpha = 0.1$, $\beta = 1$

L_{seg}

Segmentation loss: distance between the target labels and the predicted labels

L_{sem}

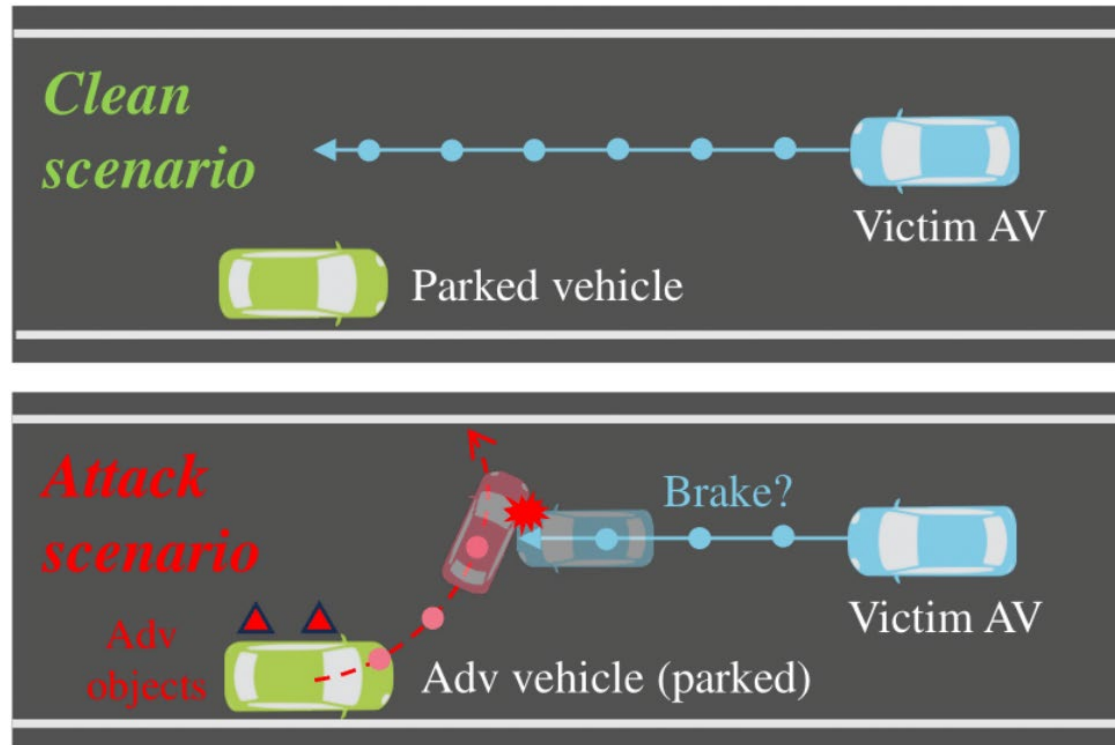
Semantic loss: keep features of the unrelated scene unchanged (stealth)

L_{occ}

Occlusion loss: prevent the object from being blocked, so the laser can hit it

- L3.2: variables = pixels in an ϵ -ball. Here: variables = **where to put physical points**; constraints come from sensing physics
- Gradients flow through a differentiable approximation of the voxelization / feature pipeline; Adam optimizer in the paper

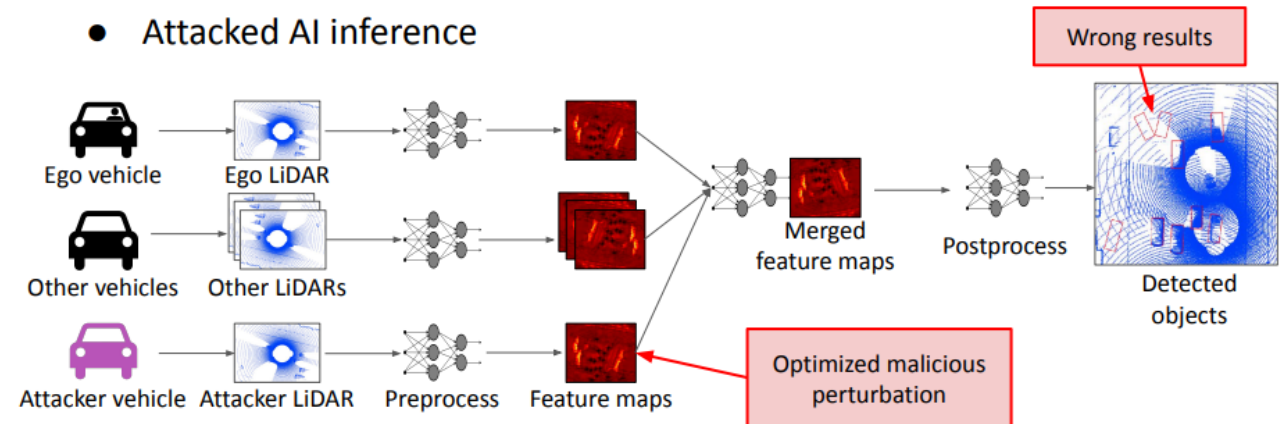
Beyond Detection: Fooling Trajectory Prediction



- Lou et al., USENIX Security 2024: LiDAR-induced deceptions cause a **false predicted trajectory** of a parked vehicle
- Adversarial objects (cardboards) trigger **unsafe avoidance**: unnecessary braking or lane change

Collaborative Perception: Fabricated Features

- Nearby vehicles share raw LiDAR or **feature maps** and fuse them: better perception in occluded scenes
- New attack surface: a **malicious vehicle fabricates data or features**, and the fused output is corrupted
- Zhang et al. (USENIX Security 2024) study attacks and countermeasures for data fabrication
- Different from the ETSI Collective Perception Message, which shares object lists



LiDAR Attacks at a Glance

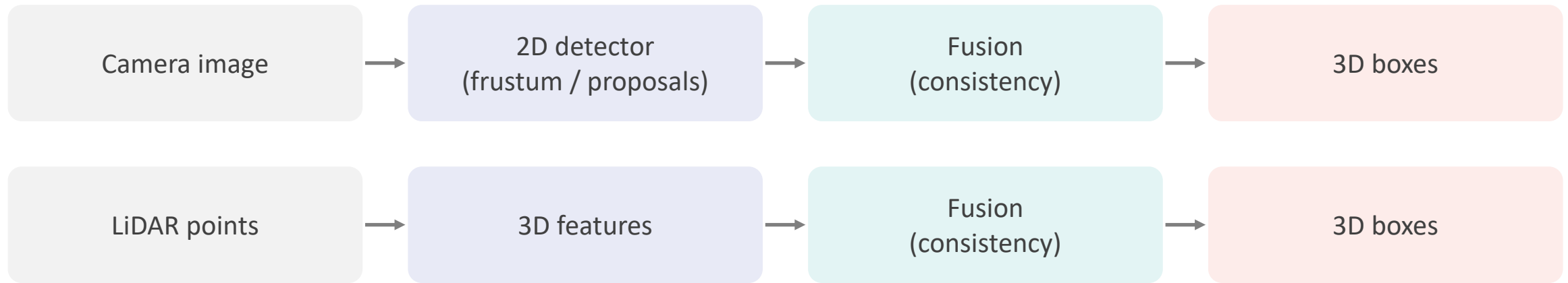
Attack (venue)	Channel	Target	Reported result (own setup)
Cao et al. (CCS 2019)	laser spoofing	Apollo LiDAR perception	≈ 75 % success
Sun et al. (USENIX Sec 2020)	laser spoofing, black-box	multiple LiDAR detectors	≈ 80 % mean success
Cao et al. (USENIX Sec 2023)	laser removal	Apollo, Autoware, PointPillars, fusion	92.7 % remove 90 % of points
Tu et al. (CVPR 2020)	adversarial object (roof)	LiDAR 3D detectors	80 % hide the vehicle
Zhu et al. (SenSys 2021)	adversarial objects	segmentation networks	> 90 % in real-world tests
Lou et al. (USENIX Sec 2024)	adversarial objects	trajectory prediction	false trajectory → unsafe avoidance
Zhang et al. (USENIX Sec 2024)	V2X feature fabrication	collaborative perception	corrupted fused detections

Numbers as reported in the cited abstracts; setups differ and are not directly comparable.

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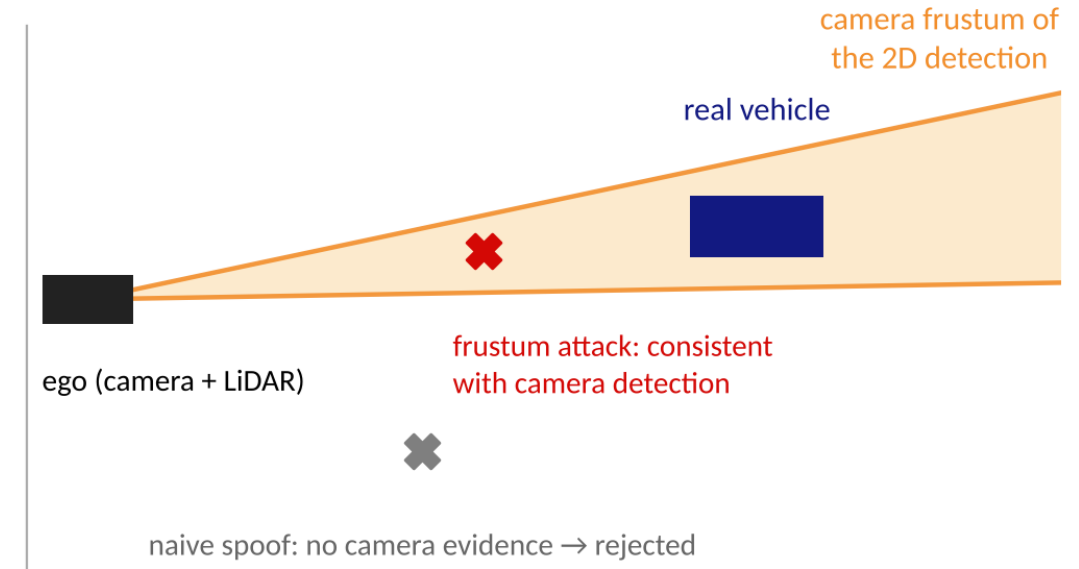
Camera-LiDAR Fusion: Why It Looked Robust



- Assumption: an attacker must fool **both** sensors **consistently** to change the output
- Naive LiDAR spoofing: injected points have no camera evidence, so fusion discards them
- Fusion pipelines use cross-modal checks: semantic agreement, geometric agreement (frustum)
- Question: can an attacker who knows these checks **satisfy** them?

The Frustum Attack (USENIX Security 2022)

- Hallyburton et al.: **black-box** attack that spoofs LiDAR points **inside the camera frustum** of an object
- It preserves **consistency between camera and LiDAR semantics**, so existing LiDAR-spoofing defenses do not fire
- Succeeds against **8 perception algorithms**: 3 LiDAR-only and 3 camera-LiDAR fusion architectures
- Repeated over time it forms **longitudinal attack sequences** that compromise **tracking** and end-to-end **AV control**

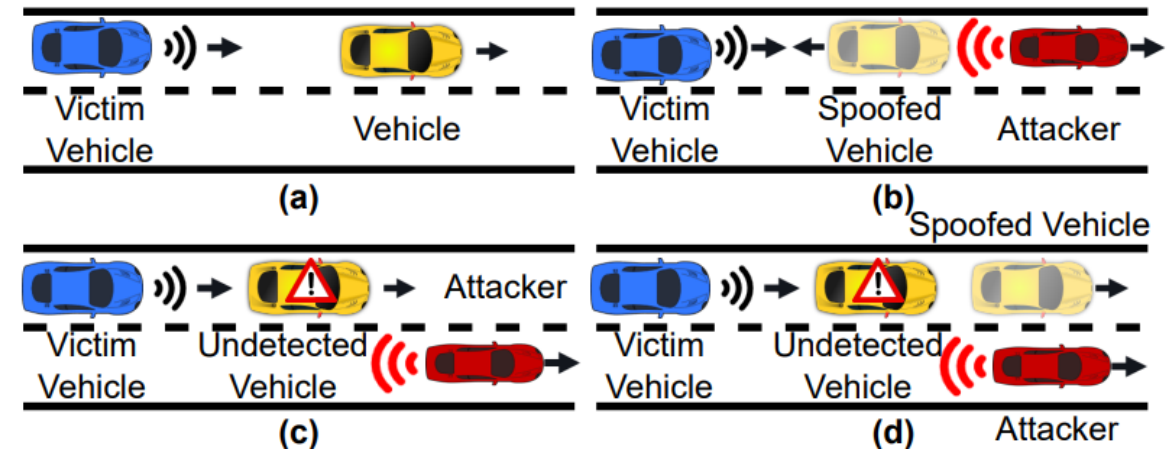


MSF-ADV: Invisible to Camera and LiDAR

- Cao et al. (IEEE S&P 2021): first study of **multi-sensor-fusion perception** under physical-world attack
- A **3D-printed adversarial object** that makes the AD system fail to detect it: invisible to camera **and** LiDAR
- Over **90 %** success across object types and MSF configurations; **100 %** collision rate in simulation against industry-grade systems
- Technical challenges solved: **non-differentiable** camera and LiDAR sensing, and **cell-level aggregated features** in LiDAR perception
- Transfers across MSF algorithms

Does More Sensors Mean More Safety?

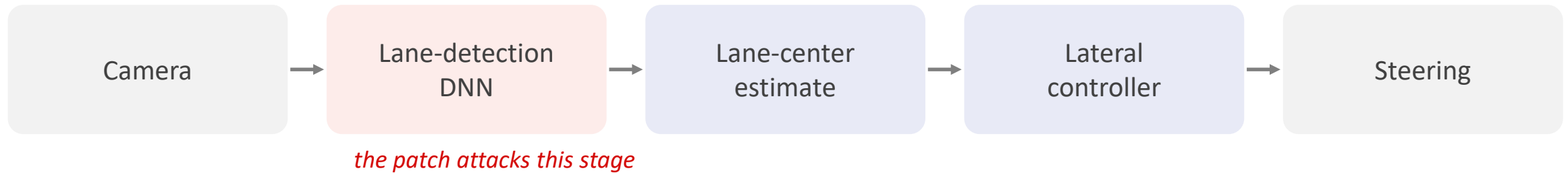
- **Yes** against independent, random failures: redundancy helps
- **Not automatically** against an attacker who understands the fusion logic: frustum attack and MSF-ADV
- Independent physics helps but is not immune: **radar** can also be attacked to add, remove or move objects (MadRadar)
- Design rule: evaluate fusion against **cross-modal adaptive attacks**, and use sensors whose failure modes are truly independent



Outline

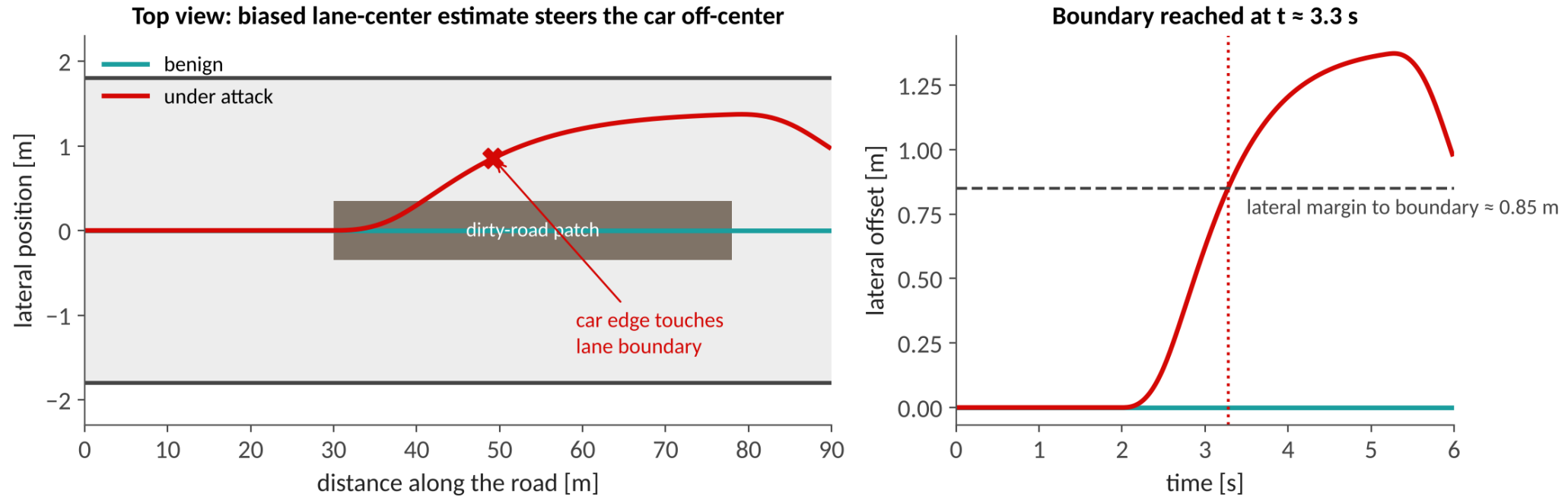
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Dirty Road Patch: Attacking Lane Centering



- Sato et al., USENIX Security 2021: first systematic study of **DL-based automated lane centering** under physical-world attack
- Attack vector: a **dirty road patch** that resembles an ordinary road stain, placed on the lane
- Optimization-based design; domain-specific challenges: **camera-frame dependence** (the patch moves in the image as the car drives) and the lane-detection **objective**
- The harm is defined on the **vehicle's lateral deviation**, not on a classification label

Why a Small Bias Can Steer a Car



- Toy model: 15 m/s, lane half-width 1.8 m, car half-width 0.95 m $\Rightarrow$ lateral margin ≈ 0.85 m; the patch biases the perceived lane center by 1.4 m
- The controller faithfully tracks the wrong center: the car crosses the boundary after ≈ 3 s, while the benign run stays centered
- Lesson: define the attack objective on **control error**, and evaluate with a vehicle model in the loop

Dirty Road Patch: Results

> 97.5 %

attack success rate

< 0.9 s

average time to success
(0.903 s)

100 %

collision rate in
software-in-the-loop + real
vehicle

80

scenarios from real
driving traces

Robust

Across lighting conditions, view angles and model designs; still effective with automatic emergency braking on; visually inconspicuous to drivers

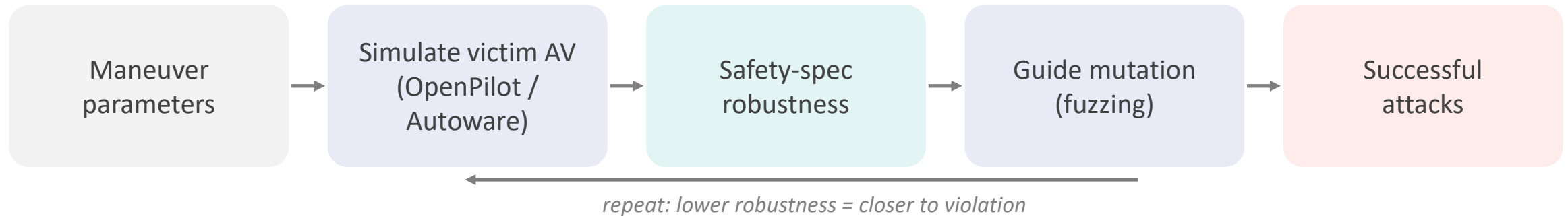
Discussion

Success time is shorter than typical driver reaction time. What should a safe system do when the lane estimate and the map disagree?

Adversarial Driving Maneuvers: Acero

- Song, Ozmen, Kim, Muller, Celik, Bianchi (Purdue): attack the AV with **normal-looking driving**, not with pixels or lasers
- **Threat model**: an attacker vehicle performs realistic maneuvers; nothing is modified in the environment
- **Goal**: force the victim AV to violate safety rules and cause physical consequences (collision with vehicles, pedestrians, objects)
- Formalizes **7 safety requirements** and **7 physical constraints** (the attacker must not endanger itself or break traffic laws)
- Targets the **planning / control** layers; perception defenses do not help

Acero: Robustness-Guided Fuzzing



- Safety rules (e.g., time-to-collision, staying in lane) are written as **temporal-logic** specifications with a quantitative robustness value
- Physical constraints keep the attacker legal and safe: the search space is **realistic driving**, not arbitrary trajectories
- **Trajectory-similarity clustering** groups successful attacks into distinct root causes
- Same idea as software fuzzing, with a **physical, closed-loop oracle**

Acero: Results

219

attacks found
OpenPilot

122

attacks found
Autoware

73.3 %

of attacks end in a collision
(vehicles, pedestrians, objects)

Takeaway

No pixel, point or signal was modified.
Perception-focused defenses (adversarial training, sensor checks) are irrelevant to this attack.

Caveat

Results are from simulation against specific AD software; they describe the failure category rather than a real-world rate.

What Each Paper Demonstrated

Paper	How it reaches the vehicle	Strongest evidence reported
Ma et al., tracking (VehicleSec 2023, WIP)	dynamic patch on monitor / projector	Level 1-2: tracking errors, 90 % / 80 % (preliminary)
Cao et al., spoofing (CCS 2019)	laser spoofing	Level 1 ($\approx 75\%$) plus two real-world scenarios
Sun et al. (USENIX Sec 2020); Cao et al. (USENIX Sec 2023)	laser spoofing; laser removal	Level 1 ($\approx 80\%$; 92.7 %)
Tu et al. (CVPR 2020); Zhu et al. (SenSys 2021)	adversarial objects	Level 1 (80 %; > 90 %)
Hallyburton et al., frustum (USENIX Sec 2022)	laser spoofing in camera frustum	Level 1-3: tracking and end-to-end control outcomes
Cao et al., MSF-ADV (S&P 2021)	3D-printed object	Level 3: 100 % collision in simulation
Sato et al., dirty road (USENIX Sec 2021)	road patch	Level 3: simulation + real vehicle, 100 % collision
Song et al., Acero (USENIX Sec 2023)	driving maneuvers	Level 3: 73.3 % of attacks end in a collision

Levels assigned from the abstracts and talk slides of the cited papers; check each paper for details.

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Defending AD Perception: A Layered View

Layer	Defense idea	Example from this lecture	Limitation
Sensor	harden the sensing process (e.g., timing randomization, filtering)	countermeasures proposed with spoofing / removal attacks	hardware changes; adaptive attackers
Model	adversarial training, augmentation, robust architectures	SVF ($\approx 2.3\%$ success); augmentation vs. adversarial objects	attack-specific; costs accuracy and latency
Physical consistency	check invariants of sensing physics	CARLO: occlusion pattern ($\approx 5.5\%$ success)	bypassed if the attacker models the check (frustum attack)
Cross-sensor / temporal	compare modalities and frames	camera-LiDAR consistency; track plausibility	defeated by consistent attacks; need diverse sensors
System	safe fallback, monitors at planning, adversarial testing	fuzzing of planners (Acero) to find unsafe behaviors	needs formal safety rules and simulation coverage

Evaluating Attacks and Defenses Honestly

Question to ask of an attack paper	Why it matters
What is the threat model (channel, knowledge, cost)?	A laser spoofer and a cardboard box are very different threats
Is it physically realized, or digital only?	Digital success rarely transfers: viewpoint, lighting, sensor noise
What is measured: per frame, per sequence, or closed-loop?	Level 1 success does not imply level 3 harm (tracking, controller)
Which stack and version?	Results on Apollo / Autoware / OpenPilot depend on versions and configuration
Is the defense evaluated against an adaptive attacker?	Non-adaptive evaluation overstates robustness
Are the claims preliminary (WIP)?	Quote success rates with their limits, not as universal risk

Discussion Questions

- Camera-only vs. LiDAR + camera + radar: which is easier to attack, and what does 'easier' mean (cost, access, stealth)?
- A one-frame detection failure: is it a safety problem? When does it become one (tracking, controller, fallback)?
- Design a defense for **removal attacks** that acts before the neural network. What does it cost?
- Acero finds attacks with legal driving. Who is responsible: the attacker, the AV maker, or the regulator?
- Pick one paper from the reading list and place it on the three-level ladder; what evidence is missing for the next level?

Next Lecture and Further Reading

- **L3.4:** not only wrong outputs but late outputs: latency and availability attacks on AD perception, tracking, and VLMs
- Curated list of AD security papers (attacks and defenses): github.com/ideasplus/Awesome-Autonomous-Driving-Security
- USENIX talk pages: Hallyburton et al. 2022 (usenixsecurity22/presentation/hallyburton); Sato et al. 2021 (usenixsecurity21/presentation/sato); Song et al. 2023 (usenixsecurity23/presentation/song)
- Project pages: sites.google.com/view/secure-safe-ai/drp-attack (Dirty Road); cpsl.pratt.duke.edu/research/frustum-attack
- Foundations: adversarial-ml-tutorial.org (Kolter and Madry, NeurIPS 2018)

References

- Cao et al., Adversarial Sensor Attack on LiDAR-based Perception in Autonomous Driving, CCS 2019
- Cao et al., Invisible for both Camera and LiDAR (MSF-ADV), IEEE S&P 2021; Cao et al., You Can't See Me, USENIX Security 2023
- Hallyburton et al., Security Analysis of Camera-LiDAR Fusion Against Black-Box Attacks, USENIX Security 2022
- Hunt et al., MadRadar, NDSS 2024; Jia et al., Fooling Detection Alone is Not Enough, ICLR 2020
- Lou et al., A First Physical-World Trajectory Prediction Attack via LiDAR-induced Deceptions, USENIX Security 2024
- Ma, Wang, Chen, Shen, Towards the Practicality of the Adversarial Attack on Object Tracking in AD, VehicleSec 2023 (WIP)
- Sato et al., Dirty Road Can Attack, USENIX Security 2021; Song et al., Discovering Adversarial Driving Maneuvers, USENIX Security 2023
- Sun et al., Towards Robust LiDAR-based Perception in Autonomous Driving, USENIX Security 2020
- Tu et al., Physically Realizable Adversarial Examples for LiDAR Object Detection, CVPR 2020
- Zhang et al., On Data Fabrication in Collaborative Vehicular Perception, USENIX Security 2024
- Zhu et al., Adversarial Attacks against LiDAR Semantic Segmentation in Autonomous Driving, SenSys 2021